

Arduino Uno Pin Connections with Components

Arduino Uno pins	Connected to Sensors	
5V pin	Connected to VCC pins of HC-SR04	
Pin 2	HC-SR04-1 Trigger Pin	Ultrasonic Sensor (US1)
Pin 3	HC-SR04-1 Echo Pin	Ultrasonic Sensor (US1)
Pin 4	HC-SR04-2 Trigger Pin	Ultrasonic Sensor (US2)
Pin 5	HC-SR04-2 Echo Pin	Ultrasonic Sensor (US2)
Pin 6	HC-SR04-3 Trigger Pin	Ultrasonic Sensor (US3)
Pin 7	HC-SR04-3 Echo Pin	Ultrasonic Sensor (US3)
Pin 8	HC-SR04-4 Trigger Pin	Ultrasonic Sensor (US4)
Pin 9	HC-SR04-5 Echo Pin	Ultrasonic Sensor (US4)
Pin 13	+Ve pin of PZ1	Piezo Buzzer (PZ1)
GND	-Ve pin of PZ1 and GND pin of HC-SR04	



Fig. 3: HC-SR04 ultrasonic sensor

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno
- US1-US4 - HC-SR04 ultrasonic sensor module

Miscellaneous:

- 9V PP3 battery
- DC 2.1mm plug
- Jumper wires
- Cap

used, so you need to define the buzzer pin in the code. Here, two variables are needed for each sensor: the duration received from the sensor and the distance to be calculated from the pulse duration.

Set the input and output pins of the four sensors and the buzzer. Also set the serial communication for monitoring the working of the four HC-SR04 sensors. Now in the loop, start initialising the trigger pin of the sensor to the high state for ten microseconds, and then using `PulseIn()` function read the output from the sensor and

calculate the distance in centimetres. The initialisation is similar for all four HC-SR04 sensors.

There are four variables: `distance1`, `distance2`, `distance3`, and `distance4` to mark the distance from all four directions. Perform logical 'or' operation. Next, use an 'if' statement to compare with the safe distance and turn on the buzzer if the distance crosses the threshold limit.

Construction and testing

Before assembling the circuit, make sure the source code (`social_distancing.ino`) is uploaded to the Arduino board using Arduino IDE.

As shown in the prototype (Fig. 1), all the four HC-SR04 sensors along with Arduino are placed on the cap in all four directions. To power the circuit, a 9V PP3 battery is connected to the Arduino through its DC jack.

For testing purposes, a safe distance of 30cm is used in the program. It can be changed as per the norms. The serial monitor utility in Arduino IDE can be used to check the distance from all four sensors. You can remove the cable after verifying the working of each sensor on the serial monitor.

In order to reduce the bulkiness of the circuit, you can use smaller ultrasonic sensor modules and replace the Arduino Uno with an Arduino Nano board. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



K. Murali Krishna is currently working as a junior engineer (telecom), BSNL, Rajahmundry, Andhra Pradesh. He is an electronics enthusiast, circuit designer and technical articles writer

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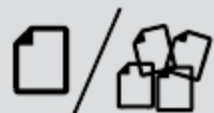
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